Lab 0: Top Level

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***Abstract*—**

I. Introduction

A. Objectives: What We Are Trying to Do

This lab builds and applies a complete discrete-time queueing simulation platform, from raw workload generation through empirical analysis and multi-scale modeling. The platform is organized into six objectives. Objectives A through C establish the toolchain: Objective A generates synthetic Poisson-arrival workloads; Objective B implements a configurable multi-server FIFO queue simulator supporting arbitrary service-reduction operations (`Sub-X`); and Objective C provides nine analysis and visualization tools that measure utilization, saturation, and errors following the USE method [**?**], along with response-time percentiles (P10, P50, P90). Objective E extends the single-queue engine into a general queueing network, routing jobs across multiple bounded queues connected by a directed graph.

B. Motivation: Why It Is Important

II. Theory

III. Mechanisms

IV. Analysis and Results

V. Summary

References